



Rajiv Gandhi Institute of Petroleum Technology, Jais Amethi

Department of Computer Science and Engineering

Object-Oriented Methodologies Lab (CS224L)

Assignment 2

Q.1 Write a C++ language program to calculate the cost of carpeting a rectangular room in a house. The program must have the following:

- Request the length and breadth of the room (assume they are in meters) from the user interface by displaying the message on the user screen.
- Applying proper constraints on the received input. If the user enters a negative value, then display the appropriate message and stop the execution of the program until the user did not enter the correct data.
- Request the cost per square meter of the carpet from the user interface by displaying the message on the user screen.
- Calculate the area of the room.
- Calculate the cost of the carpet for the room.
- Print the area and the cost.

Q.2 Write a C++ program to get 3 trials to guess a randomly generated number. Below are the hints:

- If the guessed number is bigger than the actual number, the program will respond with the message that the guessed number is higher than the actual number.
- If the guessed number is smaller than the actual number, the program will respond with the message that the guessed number is lower than the actual number.
- If the guessed number is equal to the actual number or if the K trials are exhausted, the program will end with a suitable message.

Q.3 Write a C++ program that generates a sequence of integers from a given positive integer n, as follows.

- If n is even, divide it by 2.
- If n is odd, multiply it by 3 and add 1.
- Repeat this process with the new value of n
- Stopping when n = 1.
- For example, if n is 13, the following sequence will be generated:

13 40 20 10 5 16 8 4 2 1